AdaBoost

AdaBoost short for **Adaptive Boosting** is a machine learning method that combines many simple models (like tiny decision trees) to make a stronger model.

- It trains one model at a time.
- After each round, it pays more attention to the data points it got wrong.
- It keeps combining these models, giving more weight to the better ones.
- In the end, it creates a powerful model that is good at making predictions.

AdaBoost helps improve accuracy and reduce errors in both classification and regression tasks.

AdaBoost - How it works

1. **Initialization**: Equal weights are assigned to all training instances.

2. **Iteration:**
   - A weak learner is trained on the weighted data.
   - The learner's error is calculated, and its influence on the final prediction is determined.
   - Weights of misclassified instances are increased, making them more important for the next iteration.

3. **Combination:** The outputs of all weak learners are combined using a weighted sum or voting.

AdaBoost

Trainer
weak classifier 1
weak classifier 2
weak classifier K
Weighted Voter
Classifier
Non-linear classifier
Feature 2
Feature 1

AdaBoost

Advantages

☑ Improved Accuracy

☑ Reduced Overfitting

☑ Handles Noisy Data

Disadvantages

☒ Sensitive to Outliers

☒ Can Overfit

☒ Computationally Intensive

AdaBoost

Applications

Anomaly detection

Object Recognition

Face Detection

Credit scoring

Text Classification

XGBoost

XGBoost stands for **Extreme Gradient Boosting**. It's a fast and powerful machine learning method used to make very accurate predictions.

- It builds many small decision trees one after another.
- Each new tree fixes the mistakes made by the previous ones.
- All trees work together to give better results.
- It works great for both classification (like yes/no) and regression (predicting numbers).
- XGBoost is very popular because it's fast, handles big data, and often wins ML competitions.

XGBoost - How it works

- **Start with a Simple Prediction**
  Begin by predicting the average value of the target variable.

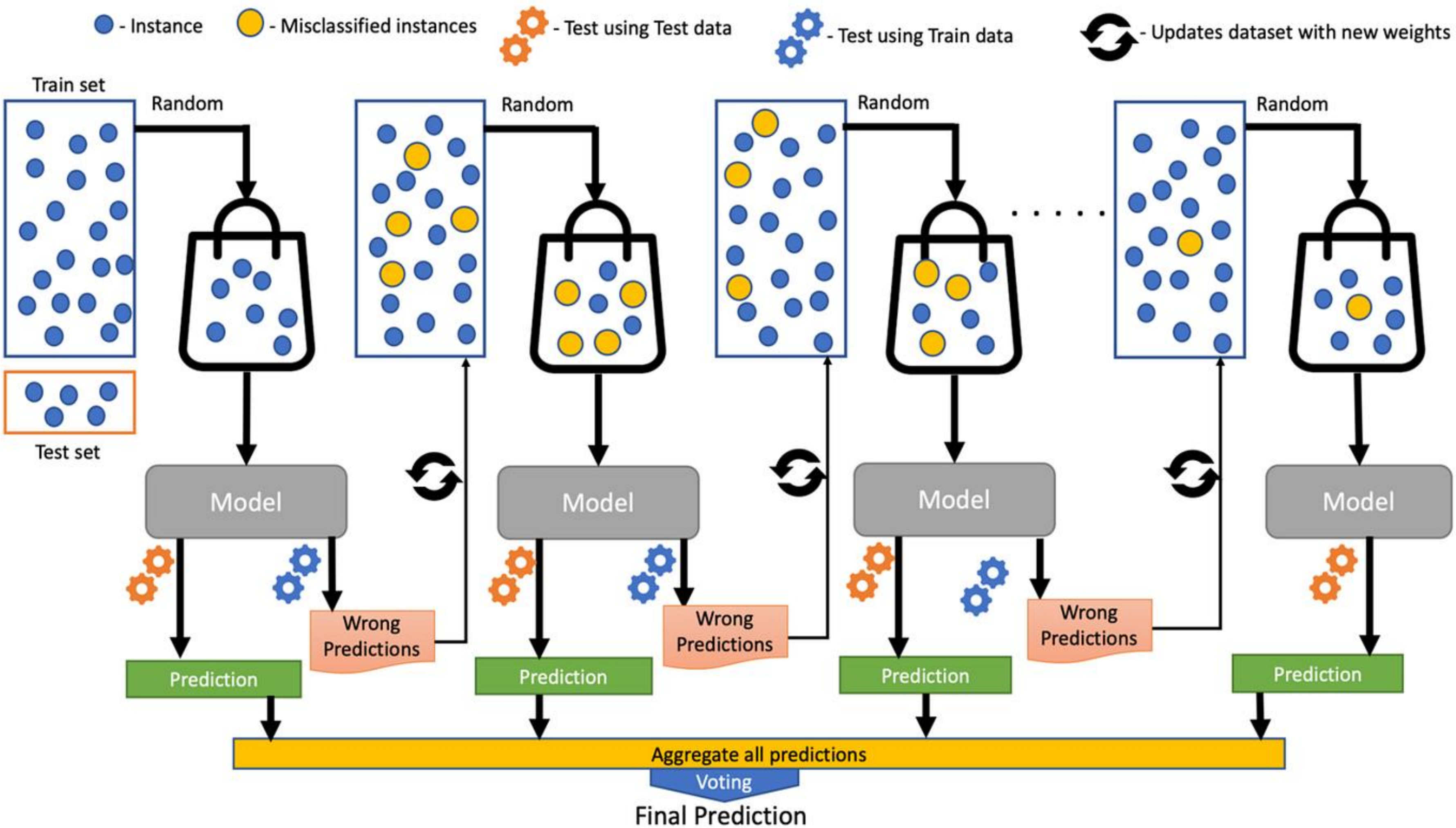
- **Find the Errors (Residuals)**
  Calculate how far off the predictions are from the actual values.

- **Build a Tree to Fix the Errors**
  Create a small decision tree that learns to correct those errors.

- **Update the Model**
  Add the tree's predictions to improve the overall result.

- **Repeat and Combine**
  Repeat the process with more trees, each fixing the remaining errors.
  Final prediction = sum of all tree outputs.

XGBoost



- Instance
- Misclassified instances
- Test using Test data
- Test using Train data
- Updates dataset with new weights
Train set
Random
Test set
Model
Wrong Predictions
Prediction
Random
Model
Wrong Predictions
Prediction
Random
Model
Wrong Predictions
Prediction
Random
Model
Prediction
Aggregate all predictions
Voting
Final Prediction

XGBoost

Advantages

- ☑ High Accuracy
- ☑ Handles Missing Data
- ☑ Feature Importance

Disadvantages

- ☒ Overfitting
- ☒ Sensitivity to Outliers
- ☒ Potential for Bias

XGBoost

Applications

- Fraud detection
- Product Recomendation
- Loan default prediction
- Credit scoring
- Spam filters

LightGBM

LightGBM **Light Gradient Boosting Machine** is a fast and powerful machine learning tool used for making predictions.

- It builds decision trees very efficiently, even on large datasets.
- It uses a smart method called leaf-wise growth to improve accuracy.
- It includes special techniques like GOSS and EFB to speed up training and reduce memory use.
- LightGBM often works faster and better than other boosting models like XGBoost.

LightGBM - How it works

- **Build Trees One-by-One (Gradient Boosting)**

  Build decision trees in sequence. Each tree fixes the errors made by the previous ones.

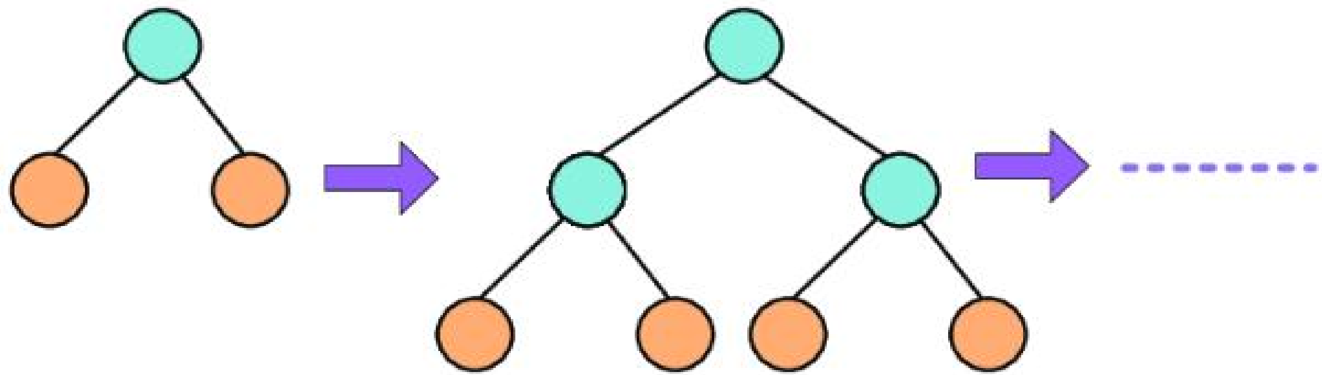
- **Use Leaf-Wise Tree Growth**
  Choose the best leaf (with most error) to split.

- **Convert Data into Histograms**
  Turns feature values into bins (like buckets) to speed up training.

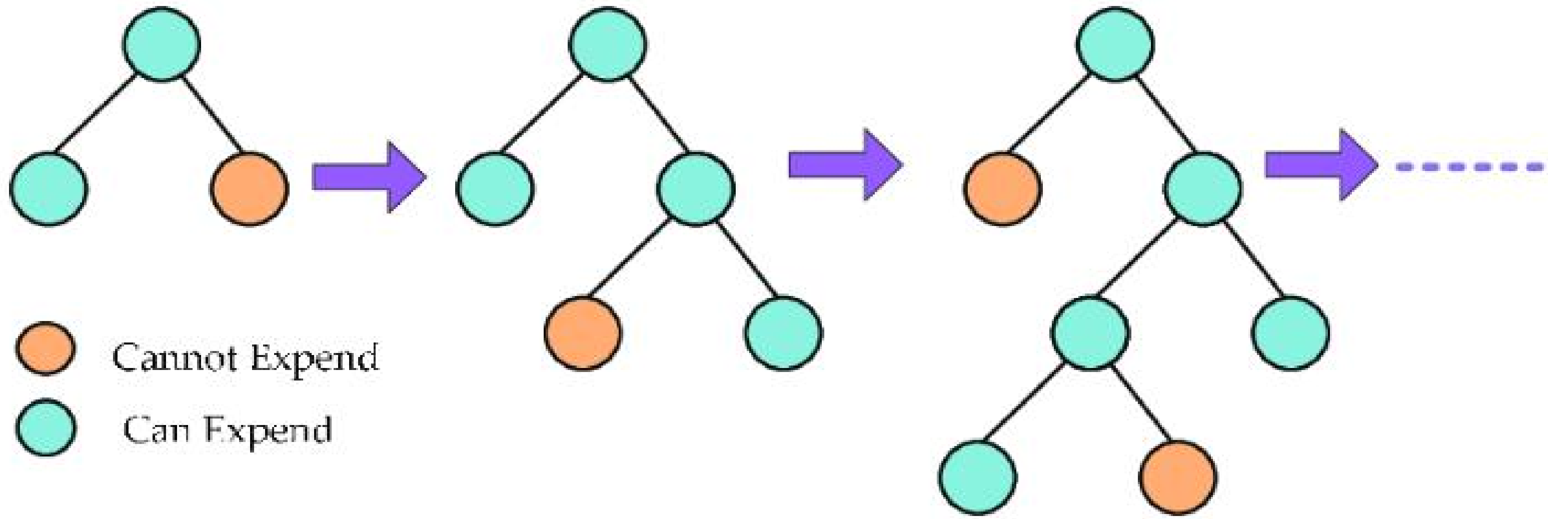
- **Focus on Important Samples (GOSS)**
  Gives more attention to data points with bigger prediction errors.

- **Bundle Features (EFB)**
  Combines features that don’t overlap into one. Reduces number of features, boosting speed.

LightGBM



Level-wise tree expansion



Leaf-wise tree expansion in LightGBM

LightGBM

Advantages

- ☑ Lower Memory Usage
- ☑ Large Dataset Compatibility
- ☑ Handles Categorical Features - no Preprocessing

Disadvantages

- ☒ Sensitivity to Small Datasets
- ☒ Less Robust to Outliers
- ☒ Potential Overfitting

LightGBM

Applications

- Fraud detection
- Large Scale Recomendation
- Sales Forecasting
- Credit risk scoring
- Click-Through Prediction